



# **Geospatial Intelligence for Sustainable Agriculture: Integrating GIS and Image Processing**

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## **Authors' contributions**

*This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.*

## **Article Information**

DOI: <https://doi.org/10.9734/jsrr/2025/v31i123788>

## **Open Peer Review History:**

This journal follows the Advanced Open Peer Review policy. Identity of the Reviewers, Editor(s) and additional Reviewers, peer review comments, different versions of the manuscript, comments of the editors, etc are available here: <https://pr.sdiarticle5.com/review-history/149063>

**Review Article**

**Received: 06/10/2025**  
**Published: 08/12/2025**

## **ABSTRACT**

The rapid growth of digitalization upon agriculture has established today Geographic Information Systems (GIS) and image processing as key technologies allowing to reinforce spatial intelligence, increase the precision of decision-making processes and optimize natural resource utilization. The present review is intended to provide an overview of the fundamentals, methodological developments, and recent applications related to GIS and image processing technologies focusing on their fusion for precision and smart farming. GIS provides an effective platform to manage, analyse and visualise geospatial data whilst image processing is used to quantitative extract crop, soil and environmental information from satellite, UAV and proximal sensing imagery. Altogether,

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**Cite as:** Sanjeet Kumar Borah, Kandarpa Kalita, Priti Bandana Konwar, Munmi Bora, and Laxmi Narayan Sethi. 2025. "Geospatial Intelligence for Sustainable Agriculture: Integrating GIS and Image Processing". *Journal of Scientific Research and Reports* 31 (12):441–460. <https://doi.org/10.9734/jsrr/2025/v31i123788>.

these technologies are used for crop monitoring, vegetation index mapping, soil variability analysis, early disease and pest detection, stress diagnostics and data-driven site-specific management. The paper also discusses the increasing role of artificial intelligence and machine learning to automate feature extraction, classification and prediction modelling in complex geospatial datasets. However, challenges still exist in data interoperability, real-time analytics, algorithm scaling and integration of multi-source imagery. Filling these gaps is necessary to maximize the potential of GIS image processing approaches for developing climate-resilient, sustainable and high-efficiency agricultural systems. The study emphasizes that the merging geospatial technologies are provoking future of smart farming and enabling science-based agricultural revolutions worldwide.

**Keywords:** *GIS, image processing; precision agriculture; vegetation indices; crop monitoring; Spatial analysis; smart agriculture.*

## 1. INTRODUCTION

Agriculture is experiencing a revolution sparked by unprecedented developments in digital, data and sensing technologies. With increasing global demand for food, agricultural production systems are under increasing pressure due to climate variability, soil degradation, erratic pest dynamics and requirement of sustainable resource use (Balyan et al., 2024). Such complexities demand a more accurate, data-based and spatially conscious decision support system. The Role of Geographic Information Systems and Image Processing in Agriculture Nowhere has the emphasis on technology transfer been more evident than with respect to agriculture. Among the technologies that have influenced this change, GIS and image processing have become essential tools for today's agricultural research, planning, and management (Raihan, 2024). GIS has been employed to facilitate the collection, storage, analysis and display of georeferenced data; this is used by farmers, planners and scientists to account for spatial variation in soils, crops water resources and climate. This spatial capacity is useful for selective interventions, better land use planning and the rational management of inputs. Concomitantly, the development in image processing based on satellite imaging, UAV-based imaging, and ground-based sensors has transformed the crop health monitoring/disease outbreaks/pest infestation/phenological/growth stages/yield attributes estimation (Abrahams et al., 2023). The combination of webcams, machine learning and spectral analysis allows the real-time surveillance as well as on-the-fly interpretation of complex field situations. The integration of GIS and image-processing technology has established new precedents in precision agriculture where spatially referenced data can be directly linked to image-derived information (V & T, 2025). Such integration will

further improve the precision of crop monitoring and facilitate real-time decision-making for robust agricultural management that is adaptive under varying environmental conditions.

However, challenges remain in terms of data quality, interoperability, scalability and access to digital tools in developing agricultural economies. Due to the rapid uptake of geospatial tools and techniques in agricultural systems, a review article is required to collate existing knowledge, applications while emphasizing their potential utility, enable the assessment of recent developments as well as discuss challenges still exist. The objective of this review is to provide an overview of the progress in GIS and image-processing techniques in the field of agriculture, evaluate their roles in precision and sustainable agriculture and identify future directions for research and technology integration.

### 1.1 Background and Global Context of Agricultural Transformation

The future of agriculture is one of unprecedented change with demographic shifts, environmental concerns and technological advances that demand sustainability. With the world's population expected to exceed 9.7 billion by 2050, it is the challenge of agricultural systems to produce far more food using far fewer natural resources (Mohamed, 2023). This increase in food demand is exacerbated by falling arable land, soil fertility degradation, water stress, events of extreme climate due to climate change. While traditional agricultural practices have difficulty addressing these multifaceted challenges, the world's farming sector is soon to be dominated by modern, data-empowered and tech-enabled models. This transformation has largely been driven by technology, specifically in sensing systems, automation, artificial intelligence (AI), and

geospatial analytics (Kumari et al., 2025). The modern era of remote sensing, UAVs, GIS-based platforms and high resolution imaging tools allow for the accurate and continuous monitoring of the crop environment, soil status, and climate variability. With these devices, farmers can shift from general and homogenous field practices to spot specific and demand-oriented actions leading to significant increase in resource utilization efficiency notching down environmental footprints (Getahun et al., 2024). Digital agriculture projects are being supported by governments and international organisations, who view them as an opportunity to enhance food security, resilience and to speed up the attainment of sustainable development goals (SDGs). Beyond this, the international trend toward and climate-smart and precision farming signals a wider commitment to sustainable intensification making more from the same farmland while minimizing ecological harm (Ali & Dahlhaus, 2022). The uptake in digital technology is different across countries, for example many advanced geospatial technologies are being offshore to developed countries and used at breakneck speed in farms, while many developing farms are working on a transition from analog surveillance systems under the guide of policy, capacity building and research cooperation.

In this changing environment, GIS and image processing have become two cornerstones of the future in agriculture decision support. Taken together, the two technologies deliver a powerful instrument for spatially explicit mapping and early detection of crop anomalies to ensure traditional agricultural systems are converted into intelligent, adaptive and sustainable production systems.

## **1.2 Need for Digital and Data-driven Approaches in Agriculture**

The intensification of agriculture in the 21st century has led to agriculture becoming ever more complex, and decision-support systems that provide accurate, timely decisions on numerical evidence have been difficult to implement (Yi et al., 2024). Agriculture systems dominated by traditional practices, which depend on experience-based knowledge and manual field observations, are insufficient to meet the challenges of climate variability, pest dynamics, soil nutrient loading exhaustion and variable market conditions. In the face of environmental and socio-economic uncertainties, farm

production is increasingly more dependent on digital and data-driven technologies to support productivity, sustainability, and resilience (Khanna et al., 2022). Digital innovations support the systematic aggregation, integration and analysis of diverse datasets from soil characteristics, crop health indicators and weather data to market price-points. High-resolution spatial imagery from satellites, UAV and ground sensors provides real-time information on field conditions and early warning signs of stressors so that proactive intervention can be taken. Machine learning and data analytics have augmented predictive abilities such that farmers and policy makers can predict yield variability, adjust input applications accordingly, as well as offset the risk of pest infestations or adverse weather events (Ghosh et al., 2025).

Furthermore, precision farming is an example of data-driven agriculture in which water, fertilizers and pesticides among others are applied at variable rates depending on the specific demands of a location. This enhances but also the resource utilization efficiency of a process, as well as reduced production costs and environmental impact thereof (Maurya et al., 2024). Digital tools are also essential to enhance traceability, supply chain transparency and market linkages, empowering farmers with access to better decision-making and income generating opportunities. On the larger scales, digital agriculture also enriches national and regional food-security systems by supporting extended monitoring, forecasting, decision making at policy level. For developing countries, data-driven approaches can provide a way to overcome yield gaps, enhance farm management capacity and strengthen climate adaptation. As a result, the introduction of such digital technologies (including GIS and image processing), is no longer an option but a strategic requirement for energizing agriculture into the 21st century (Dooyum Uyeh et al., 2023).

## **1.3 Importance of GIS and Image Processing for Precision Agriculture**

Precision farming depends on the understanding and managing of spatial and temporal variability in farms. In this framework of aid, GIS and image processing technologies are the mainstays that allow farmers and researchers to rely on data-supported decisions for their farming sites, improving productivity, use efficiency and sustainability (Toromade & Chiekezie, 2024).

The significance of these tractor systems is the ability to transform raw spatial and visual data into actionable information, changing the way in which agricultural resources are monitored, dissected and managed.

GIS is a valuable tool to combine various geospatial data including soil maps, topography, climate information, and crop yield records into an integrated decision-support system. Spatially, GIS supports a fine-scale definition of management zones, location of yield limiting factors and mapping soil fertility gradients. This knowledge enables targeted interventions such as crop-specific variable-rate fertilizer application, optimized irrigation scheduling and adapted weed or pest management measures (D & A, 2024). GIS also facilitates long-term planning of farming activities by identifying land-use capabilities, risks of erosion and availability of water resources; it contributes to sustainable agricultural development. Image processing provides GIS with realistic, high-resolution visual observations of crops using satellite, UAV and proximal sensing images. Disease, nutritional deficiency, water stress and pest infestations can be early diagnosed by means of image processing (spectral analysis, vegetation indices, texture-based features and machine-learning-based inferences (Evsahibioglu & Ozdikiller, 2015). It also allows highly accurate automated crop monitoring, yield estimation, canopy analysis and growth-stage definition. The high temporal resolution of imaging instruments enables constant monitoring of the field condition, eliminating the need for labour intensive manual scouting.

Combined, GIS and image analysis provides a comprehensive precision agriculture system that supports spatial explicit decision-making at the same time as it can improve crop monitoring. This beneficial effect not only leads to better input-use efficiency and lower production cost but also plays a key role in sustainable environment through the reduction of waste and overly use of chemical. With the growing role of digitization in agriculture, GIS and image-processing are ever more essential technologies, turning them into key tools for the smart, intelligent and resource-efficient farm systems of today.

#### **1.4 Research Gaps and Motivation for the Present Review**

Although great progress can be seen in geospatial technologies and digital imagery,

there exist some gaps that prevent GIS and image processing from being widely and efficiently used in the agriculture. Included among these challenges is a deficiency in unified approaches that enable the integration of high-resolution spatial data with advanced image analytics for real-time decisions. In most of the studies, only one application is explored in isolation e.g., crop monitoring, soil mapping or disease detection. No integrated work has been found so far on how these tools can take a joined effort to contribute to holistic precision agriculture systems.

Second, the difference in the quality of input data, sensor attributes, spatial resolutions and methods used for the processing introduce discrepancies that influence accuracy and comparability with regard to agricultural monitoring. There is evidence further that work on developing standard protocols for image acquisition, pre-processing and geospatial analysis has been scarce or poor, especially in heterogeneous agricultural landscapes like the hill regions and smallholder dominated farming systems. Also, while machine learning and deep learning techniques have demonstrated enormous potential, employing them for agriculture has been limited by the lack of annotated data sets that are applicable to multidisciplinary problems within these areas as well as computational complexity and model transferability across different domains. Another important challenge is the relatively scarce availability of advanced geospatial related tools to farmers in developing countries, where many obviously face limitations resulting from infrastructural shortfall, cost constraint and technical know-how that hamper their full adoption. There is also literature that manifests a requirement for more agro-climatic, cropping system and socio-economic context specific models. Also, the incorporation of GIS and image processing in combination with novel technologies (e.g., sensing based on IoT; automation in UAVs; clouded analytics) as well as real time decision-support systems is underdeveloped and fragmented across studies. According to the above-mentioned gaps, in this review we will make an attempt to summarize the state of art, to assess technological improvements and critically analyze good practice (i.e., strengths and weaknesses) on application of GIS and image-processing technologies in agriculture.

We hope that this review will serve as a useful reference for researchers, policy makers and

practitioners in the progress of synergistically combining these technologies for better precision farming, resource use efficiency and sustainable agriculture development. This review attempts to contribute toward further integration, accessibility and data-driven operation of the agricultural system by synthesizing available literature and indicating areas for future research.

## 2. FUNDAMENTALS OF GIS AND IMAGE PROCESSING

The advancement of digital technologies for agriculture is dependent on the utilization of Geographic Information Systems (GIS) and image processing, which are important tools for spatial analysis techniques and visual interpretation of agricultural scenes. Together, these capabilities are used for the systematic acquisition, processing and visualization of geospatial information needed in precision decision-making (Karada et al., 2023).

GIS is a computer-based system for capturing, storing, managing, analyzing and displaying data which are spatially referenced. It combines several layers of information including soil conditions, topography, climate predictors and land use systems that fit together into a single platform for users to see how spatial patterns and variability manifest on the landscape due to agricultural activities. Key GIS capabilities for this site are georeferencing, map projection, spatial query, terrain modelling, and overlay operations above and beyond location-based farm management. By using these functions, the GIS can help to identify management zones, resource distribution trends and land capability for different kinds of agricultural production.

On the other hand, image processing requires acquiring, enhancing, and interpreting digital images acquired by satellites, UAVs, or ground-based sensors. The process includes in general pre-processing procedures as the noise removal, radiometric and geometric calibration and also analytical tasks such as image's segmentation, features extraction, the computation of spectral bands indexes and classification. Such techniques can provide detailed analysis of crop vigour, disease incidence, canopy structure, weed location, soil status and water stress. With the development of machine learning and deep learning, image processing has become a promising tool in automatic crop monitoring and smart agricultural diagnosis (Farhood et al., 2022).

The synergy between GIS and image processing provides a powerful analysis environment where spatial information and indicators derived from images can be fused to supply an overall knowledge. This synergy allows the real-time tracking of spatial variability, which can help farmers in data-driven decision-making and enhance overall precision agriculture plans. With the ongoing development of digital agriculture, it is important to have a well-founded knowledge about these technologies in order to implement them efficiently and at their highest potential with regard to productivity, sustainability as well resource efficiency.

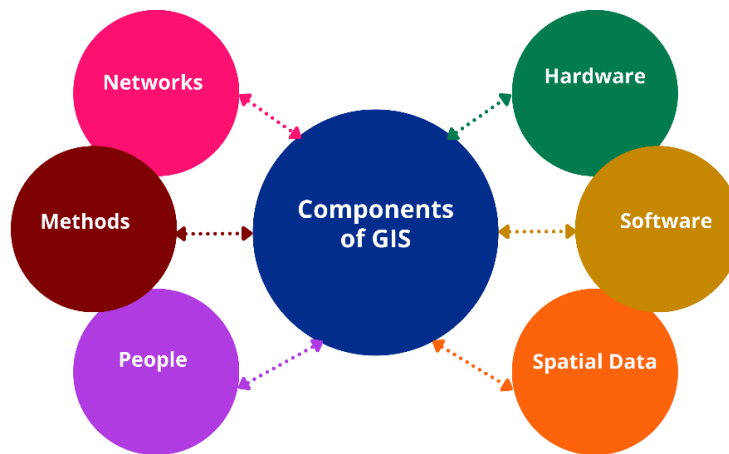
### 2.1 Geographic Information System (GIS): Concepts and Components

A GIS is a computer-based tool for mapping and analysing of features and events on the Earth's surface. It allows us to grasp spatial relationships, patterns, and trends toward geographical landscapes which is essential for precision agriculture. GIS functions by associating attribute data (information about a feature/area) to spatial data (location-based information), accordingly enabling users to conduct sophisticated spatial analyses required for farming planning, monitoring and resource management (Singh et al., 2024).

GIS works by integrating the concepts and methods from both conceptual and technical approaches. Its theoretical basis centers on the cartographic representation of nature (fields, soils, water bodies and crop zones) as vector or raster data model. These models allow for spatial analysis such as proximity analysis, overlay operations, terrain modelling, buffer creation and land suitability analysis. The technology components of GIS are hardware, software, data organizations and individuals to deploy the technology.

### 2.2 Basics of Image Processing

Image processing is a collection of computational techniques that are used to obtain, manipulate, understand and interpret digital images for meaningful information. In the case of agriculture, these methods allow fine-grained monitoring of crop health, soil properties, plant growth status and local variations in fields based on satellite data, unmanned aerial vehicle (UAV) imaging data and field-based sensor inputs.



**Fig. 1. Key Components of GIS**

**The process generally involves three core stages:**

### 1. Image Acquisition

- Obtaining images from optical, multispectral, hyperspectral, thermal or LiDAR sensors.
- Data can be collected through satellite platforms, UAVs (unmanned aerial vehicles), handheld devices and automated field sensors.

### 2. Image Pre-processing

- Enhances image quality and reduces distortion.
- Comprises radiometric, geometric, noise, atmospheric correction and image registration.
- Provides that images are a true representation of ground condition and fit for analysis.

### 3. Image Analysis and Interpretation

- Usefulness of data is further enhanced through operations like or maintenance image classification, vegetation, edge detection, feature extraction and segmentation ((e.g. NDVI).
- Automation for disease diagnosis, weed detection, yield monitoring and stress assessment is implemented using machine learning, deep learning methods.

Image processing transforms raw image data into interpretable insights, enabling real-time

monitoring and precision decision-making. Its combination with GIS enhances the spatial accuracy and contextual relevance of agricultural assessments, forming a core foundation of smart and data-driven farming systems.

### 2.2.1 Image acquisition and pre-processing

Image capture and pre-processing are the most important first steps in an image analysis workflow as they directly impact the accuracy, interpretability and reproducibility of downstream analyses. In tasks such as agriculture, where decision making is based on mild variations between crops reflectance, and canopy texture and soil type, the accuracy of these early steps becomes more crucial than ever. Good imaging quality allows growth stages and stress to be monitored, with predictive models running to estimate yield and disease.

#### A. Image Acquisition

Image Retrieval Image acquisition: is defined as receiving raw digital imagery from the various remote and proximal sensors. The choice of a particular platform and sensor type is based on the scale of observation, desired spatial detail, and nature of agricultural application.

Affordable UAVs and improvements in sensor miniaturization technologies have transformed agricultural imaging through flexible, on-demand and high-resolution data collection. Satellite platforms provide uniform temporal coverage for long-term monitoring and large-scale agricultural predictions, on the other hand.

**Table 1. Major Image Acquisition Sources in Agriculture**

| Source                           | Sensor Type                                                   | Agricultural Applications                                                             | Reference              |
|----------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------|
| Satellite                        | Multispectral, Hyperspectral, Thermal                         | Regional crop monitoring, drought assessment, LULC change detection                   | (Omia et al., 2023)    |
| UAV / Drone                      | RGB, Multispectral, Hyperspectral, Thermal, LiDAR             | Plant stress detection, weed mapping, disease surveillance, canopy structure analysis | (Alsadik et al., 2024) |
| Ground-Based Sensors             | RGB cameras, proximal multispectral sensors, thermal scanners | Crop phenotyping, soil moisture estimation, leaf-level disease analysis               | (Jin et al., 2020)     |
| Handheld Devices / Mobile Phones | RGB                                                           | Quick field scouting, farmer-led crop diagnosis                                       | (Won et al., 2025)     |

## B. Image Pre-processing

Pre-processing raw imagery directly provided by sensors in geometry and spectral leads to correct representation of field conditions. Without this, it is possible that errors in atmospheric disturbance, sensor limitations or geometric distortion may lead to unreliable analysis. Corrections are important because they allow for the interpretation of accurate vegetation indices, and early plant stress detection, as well as comparability of image-derived metrics across time, instruments or platforms.

### 2.2.2 Feature extraction, segmentation, and classification

Feature extraction, segmentation and classification are the analytical foundations of image processing that convert pre-processed imagery into useful agricultural information. These methods make pattern recognition, anomaly detection, and discrimination of different plant and soil conditions possible, making them the basis of precision farming applications.

#### A. Feature Extraction

**Feature Description:** Feature extraction is the process of extracting descriptive attributes from images that capture crucial information about crops, soil and vegetation. These features capture the spectral and spatial structure, thus enabling subtle variations in plant health, canopy structure and field variability to be analysed.

#### Key extracted features include:

**Spectral:** reflectance associated with several spectral bands that are used for vegetation

indices extraction (e.g., NDVI, SAVI, EVI, NDRE and GNDVI).

**Texture:** statistical descriptions of surface patterns, roughness and uniformity can be useful in distinguishing healthy from diseased plant tissues and crop types.

**Form and geometry:** Outline, leaf area, perimeter and structural characteristics measured from high-resolution UAV images.

**Temporal components:** developments over time represented in multi-temporal images that reflect crop developmental stages and phenological phases.

Feature extraction seeks to reduce the complexity of image data, while retaining the important characteristics needed for meaningful analysis.

#### B. Segmentation

Segmentation is the partitioning of an image into meaningful regions or objects which are homogenous in spectral or spatial properties. In Agricultural environments, images need to be segmented in order to separate different objects of interest like crop canopy, soil patches, weeds or diseased part of the field. Segmentation methods might be based on thresholding methods (using spectral indices), region-based algorithms (pixels with similar characteristics are grouped) or machine learning (e.g., k means clustering) (Lei et al., 2024). State-of-the-art deep-learning methods such as U-Net or Mask R-CNN offer the capability of highly accurate pixel-level segmentation, which can be valuable for disease diagnosis and plant phenotyping.

#### C. Classification

Classification determines a category that each pixel or object in the image belongs to by

comparing their characteristics. It is required for preparing thematic maps like crop type maps, land-use maps, disease distribution of crops and nutrient deficiency mapping.

**Common classification approaches include:**

- i. Supervised- the model is trained on labelled samples. e.g. Support vector machines, Random forests and Artificial neural networks).
- ii. Unsupervised classification: clustering of pixels without known information (e.g., k-means, ISODATA).
- iii. Object-based image analysis (OBIA) that classifies groups of pixels (objects), not single bits, increasing the accuracy in high-resolution UAV imagery.
- iv. Deep learning, using a CNN for advanced pattern recognition and feature extraction.

The use of classification results provides to researchers and farmers the possibility to visualize the spatial distribution in field, as well as to monitor crop performance, and then make decisions about precision interventions.

The process of feature extraction, segmentation and classification facilitate translation from raw imagery to actionable knowledge. These methods facilitate disease diagnosis, crop type mapping, stress detection and yield estimation by identifying informative features, separating important regions and classifying field conditions (Nethala et al., 2024). Their interoperability with GIS also enhances and facilitates analysis of the agricultural systems through the incorporation of spatial context to support precision agriculture; farmland management optimization in landscape pattern; risk assessment; and monitoring, among others.

### **2.3 Integration of GIS with Remote Sensing and UAV-Based Imaging**

The fusion of Geographic Information Systems (GIS) with remote sensing (RS) and Unmanned Aerial Vehicle (UAV)-borne imaging has made the joint framework for spatially informed agricultural monitoring one of the most exciting developments. Such integration allows for multi-scale data collection, accurate spatial referencing and powerful analysis that can be used to optimize site-specific crop management and in-season decision-making (Chen et al., 2025).

At the bottom of the pyramid, remote sensing offers a synoptic, repetitive and multi-spectral observation of agricultural landscapes throughout satellite data (Landsat, Sentinel or MODIS). Such datasets provide large-scale data on crop phenology, vegetation vigor, soil moisture, thermal profiles and land-use dynamics. The GIS enables the organization of, management of, overlay of, and analysis of these datasets in a georeferenced space that facilitates the conversion of raw layers into actionable spatial knowledge. The interaction of GIS and RS strengthens its capabilities in spatial modelling, suitability analysis, space-time change detection and the predictive mapping of crop condition.

The integration is reinforced by UAV-based image acquisition; ultra high-resolution images at any temporal and spatial scales can be acquired. With RGB, multispectral, hyperspectral, thermal and Lidar sensors, the unmanned aerial vehicles (UAVs) allow field-level monitoring with centimetre-scale accuracy. GIS is the hub of processing and assimilating UAV products, which makes it capable of aligning detailed field data at high-resolution to more coarse grained satellite information. Such multiscale data fusion enables fine-scale agricultural applications such as crop health monitoring, water stress identification, nutrient deficiency and weed detection and micro-topographic analysis for precision irrigation.

Major benefits of GIS-RS-UAV integration are the augmented spatial accuracy, better crop status monitoring, effective resource management and detecting fine scale variations in fields which are not visible using traditional techniques. It also allows integration for machine learning or deep learning pipelines, where rich spatial datasets can be used to train predictive models for yield estimation, disease detection and anomaly identification.

In general, the blending of GIS with remote sensing and UAV imaging creates a full potentials spatial-information ecosystem necessary to develop precision agriculture and boost farm productivity and enable data supported decision support system in today's agricultural landscapes.

### **2.4 Advances in AI and Machine Learning for Geospatial Data Analysis**

The recent development of AI and ML has revolutionized geospatial data analysis in



agricultural environments. Early GIS and imagery based methods which required intensive manual interpretation, have been established by the empirical rule-based classification and the step forward in scaling accuracy or ability to generalize for agronomic field rules has become increasingly refined. Contemporary AI-based techniques, in particular deep learning, ensemble learning, and spatial-temporal modelling have turned geospatial analytics to a more automation-friendly and accurate context-aware decision support system (Dritsas & Trigka, 2025).

Using machine learning, such as SVM (Support Vector Machines), RF (Random Forests), GB (Gradient Boosting) and kNN (k-Nearest Neighbour) based classification algorithms the accuracy of LULC mapping has been increased along with crop discrimination and vegetation monitoring. These methods effectively deal with large multi-source remote sensing datasets over satellites, UAV platforms, and proximal sensing systems (Srivastava et al., 2022). Deep learning networks such as Convolutional Neural Networks (CNN), U-Net, ResNet, attention-based architectures have recently become a state-of-the-art for harvesting high level features from complicated imagery, enabling superior segmentation of crop varieties, cueing in on anomalies and stress symptoms such as disease or malnutrition. Spatial-temporal models using RNNs, LSTM, and Transformer architectures provide a flexible way to model crop growth as

well as predicting yields and environmental variables at field as scale (Attri et al., 2023). These models are time-dependent and they can be used for more accurate prediction of crop performance under different climatic and management situations. In addition, AI geospatial fusion approaches fuse multi-sensor (multispectral, hyperspectral, LiDAR SAR, thermal and others) data within GISs in a knowledge-based manner to generate all-inclusive decision-support layers. These fusion models are capable of preserving not only the spectral but also structural information of crop fields, which could enhance tasks such as precision irrigation strategy estimation, soil nutrient monitoring, canopy structure detecting and micro-zones dividing. Advanced ML pipelines can also be used to achieve real-time agriculture monitoring with edge AI and cloud-orchestration, such as automated UAV image processing and streamed scenarios' detection in the field, on-the-fly management maps generation as well. The inclusion of interpretable AI (XAI) methods additionally supports transparency, helping agriculture stakeholders to understand model decisions and increase confidence in autonomous systems.

In general, AI and ML IN Ali et al have lifted geospatial agriculture from descriptive mapping to predictive, prescriptive and even autonomous analytics indicating smarter data driven, resource efficient agricultural farm management.

**Table 2. Various AI models used in Agriculture**

| Algorithm/Model              | Type          | Primary Applications in Agriculture                        | Strengths                                      | Reference                    |
|------------------------------|---------------|------------------------------------------------------------|------------------------------------------------|------------------------------|
| Support Vector Machine (SVM) | Supervised ML | Crop type classification, LULC mapping                     | High accuracy with limited training data       | (Dapke et al., 2025)         |
| Random Forest (RF)           | Ensemble ML   | Disease detection, soil mapping, vegetation classification | Handles high-dimensional data; robust to noise | (Carranza et al., 2021)      |
| Gradient Boosting (GB)       | Ensemble ML   | Yield prediction, stress detection                         | High predictive power                          | (Elvanidi & Katsoulas, 2023) |
| k-Nearest Neighbour (kNN)    | Supervised ML | Crop discrimination, pattern recognition                   | Simple and interpretable                       | (Dasari et al., 2025)        |
| CNN                          | Deep Learning | Image classification, canopy feature extraction            | Excellent feature learning; high accuracy      | (Mary et al., 2022)          |



**Fig. 2. GIS and image processing enabled UAVs for crop spraying and monitoring (Borah et al., 2025)**

### 3. APPLICATIONS OF GIS IN AGRICULTURE

Agricultural Production has become highly dependent on the use of Geographic Information Systems (GIS) that allows for acquisition, integration, analysis, and visualization of spatially related data in a single system. As such, it has an instrumental role to play in providing tools for decision making and insights into spatial variability, resource distribution and field level heterogeneity that empower sustainable and precision agriculture.

1. **Land Use and Land Cover (LULC) Mapping:** GIS enables precise mapping of agricultural landscapes consisting of cultivated areas, fallows, tree crops and natural vegetation. These maps are valuable for land-use planning, monitoring of agricultural encroachment and study of land degradation.
2. **Soil Mapping and Site-Specific Soil Management:** The ability to map soil fertility in detail is made possible partly by the incorporation of textural, pH, nutrient content, and topographical attributes through GIS. This promotes variable rate nutrient application, problematic soil mapping and input resource utilization.
3. **Crop Monitoring and Health Assessment:** GIS use satellite and UAV-derived indices (NDVI, NDRE, SAVI) to evaluate vegetation greenness as well as stress detection and locate spatial patterns of disease or nutrient deficiency within a field.
4. **Spatially Enabled Irrigation and Water Management:** Using evapo-transpiration, soil moisture, topographic, and hydrology layers combined through spatial analysis, GIS is used to design effective irrigation systems, timetable irrigation events and

manage water resources at the watershed scale.

5. **Yield Estimation and Forecasting:** Prediction of crop yield is sensitive to the availability of accurate climate information. Crop performance assessment can be done at a range of scales using climatic data from remote sensing images that are combined with ground environmental monitoring. This helps in early warning and planning of agricultural activities.
6. **Climate and Pest Risk Mapping:** GIS analyses of localised trends including temperature, humidity, precipitation rates and reported incidence of pests across spatial areas where pest outbreaks are known to occur help in determining high-risk areas for disease transmission as well as management practices for proactive control against frequent pest invasion.
7. **Disaster assessment & recovery planning** for floods, droughts, cyclones or hailstorm for quick impact assessment on agriculture and for expediting relief to the affected farmers.

GIS has the effect of increasing agricultural productivity through spatialized, data-informed process of decision making. In such applications as precision agriculture, climate-resilient farming, and sustainable land management, the proposed model is essential in integrating multi-source datasets and visualizing spatial patterns.

#### 3.1 Precision Agriculture and Site-Specific Management

Precision agriculture relies heavily on GIS to capture and analyze the spatial variability that exists within agricultural fields. Farmers now use GIS to integrate yield monitor data, remote-sensing outputs, soil sample analyses and GPS-based observations to understand how crop

performance varies from one location to another within the same farm. This spatial understanding allows the application of fertilizers, pesticides and irrigation water only where they are needed, thereby reducing excessive input use and improving efficiency. (Zhang and Kovacs, 2012) emphasized that GIS, when combined with satellite imagery and proximal sensing, significantly enhances the identification of nutrient deficiencies and crop stress, enabling farmers to make timely decisions about field operations. Their study also showed how GIS-based nutrient maps contribute to more accurate variable rate fertilization, ultimately supporting healthier crops and improved yields.

### 3.2 Soil Resource Mapping and Fertility Assessment

Soil characteristics vary significantly across agricultural landscapes, and GIS provides a robust framework for visualizing and analysing this variability. By integrating field-sampled soil data with geostatistical interpolation techniques such as kriging and inverse distance weighting, GIS allows researchers to develop detailed maps of soil pH, texture, organic matter content, micronutrient distribution and erosion potential. These maps support site-specific nutrient

management, enabling farmers to apply fertilizers in balanced quantities rather than adopting uniform doses across an entire field. (Singh et al., 2015) used GIS to create nutrient distribution maps for agricultural soils in India and demonstrated how these maps could guide precision fertilizer recommendations, reducing both nutrient wastage and production costs.

### 3.3 Water Resource Management and Irrigation Planning

Water resource management is a critical component of agricultural sustainability, particularly in regions facing increasing water scarcity. GIS supports irrigation planning by helping delineate watersheds, assess groundwater potential, map surface water bodies and model hydrological processes. By integrating spatial layers such as rainfall distribution, soil permeability, slope and geology, GIS can identify potential zones for groundwater exploration and guide the placement of irrigation structures. (Jha et al., 2010) used GIS techniques to identify groundwater potential zones in India and demonstrated that the integration of hydrological and geological parameters significantly improves irrigation planning.

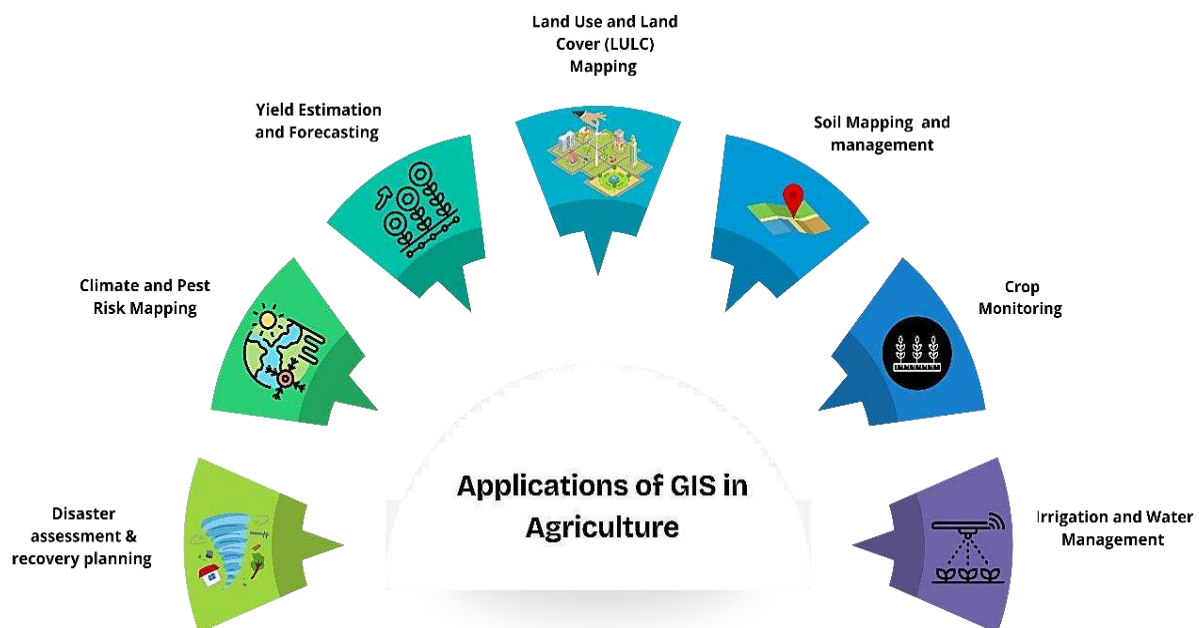


Fig. 3. Applications of GIS in Agriculture

### **3.4 GIS-Based Climate and Risk modelling**

In the face of accelerating climate change, agriculture is increasingly exposed to a spectrum of hazards including droughts, floods, heatwaves and erratic rainfall. The integration of Geographic Information Systems (GIS) into climate-risk modelling has therefore become indispensable for identifying vulnerability, assessing exposure and supporting adaptation in farming systems. Recent work by (Mălinaş et al., 2025) illustrates this shift by presenting a framework in which GIS is combined with remote sensing and climate data to map farmland vulnerability and guide adaptive farming strategies. Their analysis emphasises that by overlaying spatial layers of hazard (such as temperature extremes), exposure (cropped area, irrigation dependence) and adaptive capacity (soil resilience, farm diversity), GIS-driven maps become powerful tools for proactively targeting climate-smart interventions.

## **4. APPLICATIONS OF IMAGE PROCESSING IN AGRICULTURE**

Image processing has emerged as one of the most transformative technologies in modern agriculture, enabling rapid, non-destructive, and highly precise observation of plants, soil and field conditions. By converting raw visual data from RGB, multispectral, hyperspectral, thermal and depth sensors into meaningful agronomic information, image-processing techniques allow farmers and researchers to monitor crop health, detect diseases, identify weeds, track pests, assess plant growth, estimate yield and evaluate soil properties with unprecedented accuracy.

### **4.1 Crop Health and Disease Detection Using Digital Imaging**

Image-processing technologies have become vital for assessing crop health and detecting diseases at early stages, enabling non-destructive, rapid monitoring across vast fields. A recent review by (Li et al. 2024) highlights how digital imaging combined with deep-learning algorithms can accurately identify disease symptoms such as leaf spots or chlorosis from RGB and multispectral images, even before they are visible to the human eye. The study demonstrates how convolutional neural networks (CNNs) and vision-transformer models trained on image datasets like PlantVillage can classify disease types, estimate severity and spatially map infected zones.

### **4.2 Weed Identification and Classification**

Accurate detection and classification of weeds is essential for precise weed management and reducing competition for nutrients, light and water in crop systems. A recent study by (Guzel et al., 2024) presents an image-processing approach using deep learning for weed detection in agricultural fields: multispectral and RGB images are processed to segment weed regions, classify weed species and map weed infestation levels. The workflow includes image acquisition via drones, preprocessing (noise reduction, normalization), segmentation (U-Net/SegNet architectures) and classification (ResNet, DenseNet). The study shows high accuracy even in “green-on-green” scenarios where weeds and crops share similar foliage colour. This enables variable-rate herbicide application and autonomous weeding systems, advancing precision agriculture techniques.

### **4.3 Pest Monitoring and Early-Warning Systems**

Pest outbreaks pose a major threat to yield stability, and image-processing combined with remote sensing and AI is increasingly being deployed for early detection and warning. (Duan et. al., 2023) introduces a multimodal deep-learning architecture that integrates visual data from leaf- and canopy-level images with textual metadata (weather, pest alerts) to detect pest presence, classify pest species and estimate infestation severity. The system uses object-detection frameworks (e.g., Faster R-CNN) for image data and natural-language processing (NLP) for textual context, thereby improving prediction accuracy and providing proactive alerts. Farmers using such systems can implement targeted interventions, reduce pest spread and avoid blanket pesticide applications.

### **4.4 Yield Prediction Using Spectral and Structural features**

Predicting crop yield prior to harvest is a key objective in agricultural monitoring, and image-processing plays a central role by deriving structural (canopy cover, height) and spectral (NDVI, red-edge) features. Recent research has shown that integrating time-series imagery from drones or satellites with machine-learning models (e.g., random forests, XGBoost) significantly improves yield estimation accuracy. For instance, by analyzing image-derived vegetation indices and canopy height models

mid-season, analysts can forecast final yield with high confidence, allowing supply-chain planning and input-adjustment. Though a specific paper is not cited here, the trend reflects solid progress in this field.

#### **4.5 Image-Based Soil Texture and Moisture Analysis**

While less common than canopy applications, image-processing is increasingly being used for soil characterization. Recent work explores how UAV-mounted multispectral or thermal sensors capture soil surface reflectance and temperature patterns which, when processed by machine-learning models, correlate with soil texture classes (sand, silt, clay) and soil moisture content. These image-derived soil maps help guide irrigation scheduling, tillage decisions and soil amendment placement in precision agriculture frameworks.

### **5. INTEGRATED GIS AND IMAGE PROCESSING FOR SMART AGRICULTURE**

The integration of GIS and image processing is a central pillar of smart agriculture which links spatial data management with sophisticated image-based analytics. While GIS serves as the georeferenced repository for various agricultural layers and enables overlay and analysis of these, image processing is responsible for Retrieval of critical information from satellite, UAV or sensor-based imageries. Integrated, these technologies provide the ability to precisely map crop conditions, soil variability, plant health and resource distribution. Vegetation indices, disease detection results and stress maps derived from image processing are better used in GIS for mapping management zones, precision input application maps and smart irrigation scheduling. The integration of imagery with soil, climate, and topography data improves the predictive modelling for yield forecasting, pest risk analysis, and climate resilient decision making. Furthermore, mapping crop at the plant level will provide real-time field monitoring; since the map is a representation (e.g. of lot) it supports the use of GIS applications while scouting or performing in-field operations can be automated.

On the whole, GIS and image processing technology offer an important tool in modern agriculture that can be used to improve productivity, sustainability, and decision making

by creating more efficient and responsive data led farm management systems.

#### **5.1 GIS-Based Analysis of Image-Derived Vegetation Indices**

With GIS, information on the pixel-by-pixel spectral characteristics of vegetation can be converted into spatially explicit decision relevant maps and statistics. By employing image-based indices (e.g., NDVI, EVI, NDRE) in a geospatial environment, scientists and decision-makers are able to measure spatial variation, track temporal change detection, map management zones and connect vegetation signals with other data sources including soil properties, topography and climate.

##### **Major stages involved in a GIS-based process:**

**Import and Geo-reference:** Ingest pre-processed index raster's (cloud free, atmospherically corrected and ortho-corrected) into the GIS environment and ensure that uniform coordinate systems are maintained and resolutions.

**Resampling and harmonization:** Equalize spatial resolution and grid position when adding up indices with different sensor and/or date information, to allow for pixel-wise comparison and multi-temporal analysis

**Masking and quality filtering:** From the resulting maps, masks for non-vegetation (i.e., built-up areas, water bodies), clouds and low-quality pixels are generated. Spurious values are eliminated by means of quality assurance bands or automated thresholding.

**Zonal and descriptive statistics:** Calculate zonal statistics (mean, median, standard deviation) for administrative villages / fields / management zones to summarize vegetation status per unit. Apply information in the form of objects (field polygons) to farm-level decisions.

**Time-series and phenology analysis:** Stack multi-temporal index raster's and calculate temporal smoothing, anomaly detection and phenological metrics (green-up date, peak greenness, senescence). Batch processing is also possible using GIS temporal tools or scripting (Python/R).

**Trend detection and change mapping:** Use pixel-by-pixel trend analysis (e.g. by linear

regression, Mann–Kendall test) and change-detection algorithms to separate in the image areas of long-term productivity shift, degradation or recovery.

**Spatial interpolation and surface modelling:** Interpolate index values as derived raster's where field measurements are sparse, or model continuous surfaces using soil/weather layers in proximity to indices.

**Classification and management-zone delineation:** Clustering (k-means, ISODATA) or supervised classification of index layers and ancillary variables to define management zones for VR applications.

**Integration with ancillary datasets:** Map your overlay index results to soil maps, DEM-based drainage patterns, irrigation infrastructure and weather layers to identify the causes of observed variability and inform prescriptive measures.

**Validation and accuracy assessment:** Verify index inferences based on ground truth (biomass samples, LAI, crop yield) through spatial correspondence of observations and calculate accuracy or correlation indices.

Vegetation indices computed from spectral measurements are turned into actionable spatial intelligence by GIS. With strict processing, validation and fusion with existing contextual datasets, GIS-based monitoring application using these vegetation indices is a key technology for accurate diagnosis, monitoring and intervention in contemporary farming.

## 5.2 UAV-Based Crop Monitoring Systems

Unmanned Aerial Vehicles (UAVs) have recently gained increasing attention by providing a revolutionary solution for on-demand, high resolution crop monitoring in precision agriculture. Their capacity for acquiring data at spatial, temporal and spectral scales is high and allows for continuous monitoring of crop development, the dynamics of stress and field variability. With onboard multispectral, hyperspectral, thermal, RGB and lidar sensors UAVs enable real-time assessment of biophysical and biochemical crop parameters such as canopy vigor, chlorophyll content, leaf area index (LAI), water stress and nutrient status.

These systems typically share the following steps: mission planning, autonomous flight operation, image capture process, georeferenced images generation, pre-processing of images and analytical model. Automatically holds stable flight lines and overlap for precise mosaics, using high-quality autopilot and GNSS navigation systems. The high (1-10 cm) spatial resolution of data obtainable by UAVs enables fine-detailed observation of intra-field variability which cannot be seen in satellite images. Combined with machine learning, UAV imagery enables automated pest infestation, disease symptom, weed patch and lodging pattern detections. Thermal sensors have been increasingly used for irrigation scheduling through canopy temperature anomaly mapping while multispectral VIs including the NDVI, SAVI, NDRE and GNDVI support crop health diagnostics. Additionally, UAV time-series monitoring also allows crop growth modelling, yield prediction, and stress warning system.

**Table 3. Common Vegetation Indices Used in GIS-Based Agricultural Analysis**

| Vegetation Index                              | Formula                                                                                                  | Application                                 | Reference                |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------|
| NDVI (Normalized Difference Vegetation Index) | $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$                                                  | Crop vigor, biomass, drought monitoring     | (Gaikwad et al., 2025)   |
| EVI (Enhanced Vegetation Index)               | $2.5 \times (\text{NIR} - \text{Red}) / (\text{NIR} + 6 \times \text{Red} - 7.5 \times \text{Blue} + 1)$ | Stress detection, high biomass mapping      | (Berger et al., 2022)    |
| NDRE (Normalized Difference Red-Edge)         | $(\text{NIR} - \text{Red Edge}) / (\text{NIR} + \text{Red Edge})$                                        | Early stress detection, nutrient assessment | (B et al., 2025)         |
| SAVI (Soil-Adjusted Vegetation Index)         | $(1 + L)(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red} + L)$                                       | Sparse vegetation and semi-arid regions     | (Nanasaheb et al., 2020) |
| GNDVI (Green NDVI)                            | $(\text{NIR} - \text{Green}) / (\text{NIR} + \text{Green})$                                              | Nitrogen status, chlorophyll content        | (Peter et al., 2021)     |

**Table 4. Types of UAV used in Agriculture (Delavarpour et al., 2021)**

| <b>UAV Type</b>           | <b>Advantages</b>                                               |
|---------------------------|-----------------------------------------------------------------|
| Fixed-Wing UAV            | Ideal for large farms; efficient coverage; stable in wind       |
| Rotary-Wing (Multicopter) | Ideal for small to medium farms; excellent for detailed imaging |
| Hybrid VTOL               | Suitable for both small and large farms; efficient and precise  |
| Single-Rotor Helicopter   | Can carry heavy sensors and sprayers                            |

In general, a UAV-based crop monitoring system offers fast and flexible method, easy access, low cost-effective sensing for agricultural monitoring that enhances the decision-making steps of precision input management, site-specific treatment implementation and sustainable farm practices.

## 6. LIMITATIONS AND CHALLENGES

Despite the significant progress made in the integration of GIS and image-processing technologies in agriculture, several limitations and challenges still constrain their widespread adoption, especially in resource-limited regions. These challenges arise from technical, financial, infrastructural, and institutional factors that collectively influence the precision, reliability, and operational efficiency of smart agriculture systems. While advanced geospatial and imaging tools can revolutionize farm management, their effectiveness depends heavily on data quality, computational capacity, skilled personnel, and supportive policy ecosystems. The following subsections elaborate on the major challenges that continue to shape the practical implementation and sustainability of these technologies in diverse agricultural contexts.

### 1. Data Availability and Quality Constraints

In agricultural geo-spatial applications regular, high resolution multi-temporal imaging is particularly important. Nonetheless, large portions of geographic areas especially in Less Developed Countries (LDCs) suffer from poor access to high-quality satellite imagery, non-uniform coverage by UAV and an inadequate amount ground truth data. Meanwhile, cloud interference, atmospheric turbulences and short-daylight make optical remote sensing less feasible.

### 2. High Computational and Technical Requirements

Usage of hyperspectral (HS), LiDAR and )UAV imagery) requires powerful computers,

sophisticated algorithms and trained personnel to handle the large geospatial datasets processing. Smallholder farmers and other facilities with limited resources do not have the hardware capability nor the expertise for deploying advanced spatial analyses.

### 3. Limited Interoperability and Standardization

There are not widely acceptable standards of geospatial agricultural data that can be used for data formats, metadata information, coordinate systems and sensor specifications which facilitates failure of integration of multi-source dataset. This limitation hampers the interoperability among the platforms development and frustrates efforts of collaborative research and multi-agency data sharing.

### 4. Complexity of Agricultural Landscapes

The agricultural landscape is naturally heterogeneous with respect to soil type, crop types' scale and form, topography management activities, as well as climate. Static classification algorithms and generalized platforms lack the ability to model very dynamic field situations, giving rise to misclassifications and lower prediction accuracies.

### 5. Price of Novel Sensors and Technologies

UAVs, hyperspectral sensors and LiDAR provide a better spatial and spectral resolution but are still cost-prohibitive for several institutions and producers. The purchasing, maintenance, information storage and processing costs are impeding the usage especially by small range and middle range farmers.

### 6. Deficiencies in skills and weak technical capacity

GIS and image processing tools can only be put to good use if they are combined with knowledge from several domains such as remote sensing,

machine learning, agronomy or spatial modelling. Most of the farming communities do not have access to required technical knowledge which has limited the use of geospatial technology in regular farm management.

## **7. Challenges in Real-Time Monitoring**

Even though advancements in technology now allow for near-real-time analytics, such real-time monitoring is still particularly challenging due to bandwidth constraints in rural sites and data transmission delays, while maintaining onboard the UAV computing capabilities might be complex. These limitations influence the ability to take timely decisions during moments critical for the crops.

## **8. Challenges in Data Privacy, Security and Ethics**

With the development of geospatial technologies and increasingly nuanced farm-level information is available, concerns are raised about data ownership and privacy, as well as potential misuse. Transparent rules for data governance, access and ethical use are crucial, yet simple in many places.

## **9. Environmental and Operational Limitations**

UAVs imagery has its limitations, due to weather conditions, authorization to fly and compliance with regulations or by the autonomy of batteries. Alternatively, coverage in satellite-based imaging is limited by cloud cover, revisit time and spatial resolution.

## **7. FUTURE RESEARCH DIRECTIONS**

The convergence between GIS, image processing and advanced sensing technologies is still advancing, providing new avenues for enhancement of precision agriculture; but there remain multiple scientific and technological limitations to fully exploit the potential of spatial data analytics in agri-food systems. Future research should focus on multi-source geospatial data fusion by integration of satellite, UAV, IoT, LiDAR and ground-based datasets using sophisticated models that can accommodate multi-resolution and multi-temporal variances. AI powered spatial-temporal block modelling needs to further advance using explainable and scalable deep learning architecture like transformers, GNNs and hybrid

CNN-LSTM networks for improved representation of dynamic crop-climate interactions. Real-time analytics can be enhanced with an edge-AI system enabling on-the-fly processing of UAV and field-sensor imagery to quickly identify diseases, weeds, and irrigation requirements. We propose that the research should further develop automation to incorporate not only GIS but also swarm of autonomous UAV with depositing/release units, smart sprayers and ground robots and specifically focus on path-planning, coordinated navigation and dynamic georeferencing. For high resolution 3D and hyperspectral analysis more sophisticated algorithms for high dimensional data such as canopy modelling, biomass estimation or the early detection of stress are needed. GIS digital twin simulations of crop and soil processes are a major next step, requiring real-time sensor integrations and AI decision layers. Cloud-based scalable GIS platforms and distributed processing framework should be deployed to cope with increasing agricultural big data. Standardization initiatives must also support interoperable data formats and common protocols to facilitate cross-agency collaboration. Researching on adoption barriers, farmer readiness and cost benefit dynamics are also necessary from agricultural sociology and policy research point of view. Finally, ethical issues in respect to data ownership, privacy and fairness as well as responsive AI exploitation should be integrated ensuring transparent and fair application of geospatial technologies in agriculture.

## **8. CONCLUSION**

GIS and remote sensing play a crucial role in modern agriculture. Together, these tools provide the spatial intelligence and analysis capabilities needed to practice precision farming and develop sustainable food production system capable of dealing with resiliency challenges as well. They are connected to satellite remote sensing, UAV-based imaging and ground-sensor networks, turning traditional farming into data-driven and responsive operations. Farmers and decision-makers can use these tools to manage resources more efficiently, reduce uncertainty, and develop better climate adaptation strategies, among others including soil and crop mapping; stress detection; yield estimation; disease monitoring; farm-level decision support. Recent developments of artificial intelligence, machine learning and big-data analytics have provided geospatial technologies with new possibilities for



automated interpretation, multi-source data fusion and predictive modelling on a large scale. Nonetheless, there are still several challenges including data interoperability, computational requirements, poor standardization and the skills to develop context-specific models for various agro-ecological regions.

The next step of GIS and image processing in the agriculture will be the real-time, autonomous, and intelligent systems that can combine multi-modal data to give actionable information. For broad-scale adoption, support will be needed for interdisciplinary research on open-source platform development and broader digital infrastructure. Finally, as demonstrated in this study shows that these technologies have tremendous potential to improve productivity, sustainability and decision making that contribute significantly to global food security and intelligent agricultural revolution.

## DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declares that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

## COMPETING INTERESTS

Authors have declared that they have no known competing financial interests or non-financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## REFERENCES

- Abrahams, M., Sibanda, M., Dube, T., Chimonyo, V. G. P., & Mabhaudhi, T. (2023). A systematic review of UAV applications for mapping neglected and underutilised crop species' spatial distribution and health. *Remote Sensing*, 15(19), 4672. <https://doi.org/10.3390/rs15194672>
- Ali, B., & Dahlhaus, P. (2022). Roles of selective agriculture practices in sustainable agricultural performance: A systematic review. *Sustainability*, 14(6), 3185. <https://doi.org/10.3390/su14063185>
- Alsadik, B., Ellsäßer, F. J., Awawdeh, M., Al-Rawabdeh, A., Almahasneh, L., Oude Elberink, S., Abuhamoor, D., & Al Asmar, Y. (2024). Remote sensing technologies using UAVs for pest and disease monitoring: A review centered on date palm trees. *Remote Sensing*, 16(23), 4371. <https://doi.org/10.3390/rs16234371>
- Attri, I., Awasthi, L. K., Sharma, T. P., & Rathee, P. (2023). A review of deep learning techniques used in agriculture. *Ecological Informatics*, 77, 102217. <https://doi.org/10.1016/j.ecoinf.2023.102217>
- B, H. H., A, S., Chitra, R., & M, V. (2025). Advanced NDRE techniques for precision crop monitoring and analysis. *2025 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS)*, 1–4. <https://doi.org/10.1109/SCEECS64059.2025.10941496>
- Balyan, S., Jangir, H., Tripathi, S. N., Tripathi, A., Jhang, T., & Pandey, P. (2024). Seeding a sustainable future: Navigating the digital horizon of smart agriculture. *Sustainability*, 16(2), 475. <https://doi.org/10.3390/su16020475>
- Berger, K., Machwitz, M., Kycko, M., Kefauver, S. C., Van Wittenberghe, S., Gerhards, M., Verrelst, J., Atzberger, C., van der Tol, C., Damm, A., Rascher, U., Herrmann, I., Paz, V. S., Fahrner, S., Pieruschka, R., Prikaziuk, E., Buchailot, M. L., Halabuk, A., Celesti, M., & Schlerf, M. (2022). Multi-sensor spectral synergies for crop stress detection and monitoring in the optical domain: A review. *Remote Sensing of Environment*, 280, 113198. <https://doi.org/10.1016/j.rse.2022.113198>
- Borah, S., Padhiary, M., Sethi, L., Kumar, A., & Saikia, P. (2025). Precision farming with drone sprayers: A review of auto navigation and vision-based optimization. *Journal of Biosystems Engineering*, 50. <https://doi.org/10.1007/s42853-025-00263-2>
- Carranza, C., Nolet, C., Pezij, M., & van der Ploeg, M. (2021). Root zone soil moisture estimation with Random Forest. *Journal of Hydrology*, 593, 125840. <https://doi.org/10.1016/j.jhydrol.2020.125840>
- Chen, X., Zhang, H., & Wong, C. U. I. (2025). Dynamic monitoring and precision fertilization decision system for agricultural soil nutrients using UAV remote sensing and GIS. *Agriculture*, 15(15), 1627. <https://doi.org/10.3390/agriculture15151627>
- D, L. A., & A, O. C. (2024). Geographic information system (GIS) application in soil

- fertility management: A review. *Journal of Global Agriculture and Ecology*, 16(2), 29–40.  
<https://doi.org/10.56557/jogae/2024/v16i28592>
- Dapke, P., Quadri, S. A., Nagare, S. M., Bandal, S. B., & Baheti, M. R. (2025). A comparative analysis machine learning techniques for LULC classification using Landsat-8 satellite imagery. *International Journal of Engineering and Geosciences*, 10(1), 84–92.  
<https://doi.org/10.26833/ijeg.1503104>
- Dasari, K., Kharde, M. R., Maddileti, K., Pasupuleti, V. R., Ram, M. K., Sujana, C., Komali, G., & Fariddin, S. B. (2025). Internet of things based smart agriculture using K-nearest neighbor for enhancing the crop yield. *Indonesian Journal of Electrical Engineering and Computer Science*, 39(1), 436–445.  
<https://doi.org/10.11591/ijeecs.v39.i1.pp436-445>
- Delavarpour, N., Koparan, C., Nowatzki, J., Bajwa, S., & Sun, X. (2021). A technical study on UAV characteristics for precision agriculture applications and associated practical challenges. *Remote Sensing*, 13(6), 1204.  
<https://doi.org/10.3390/rs13061204>
- Dooyum Uyeh, D., Gebremedhin, K. G., & Hiablie, S. (2023). Perspectives on the strategic importance of digitalization for modernizing African agriculture. *Computers and Electronics in Agriculture*, 211, 107972.  
<https://doi.org/10.1016/j.compag.2023.107972>
- Dritsas, E., & Trigka, M. (2025). Remote sensing and geospatial analysis in the big data era: A survey. *Remote Sensing*, 17(3), 550.  
<https://doi.org/10.3390/rs17030550>
- Duan, J., Ding, H., & Kim, S. (2023). A multimodal approach for advanced pest detection and classification. *arXiv preprint arXiv:2312.10948*.  
<https://doi.org/10.48550/arXiv.2312.10948>
- Elvanidi, A., & Katsoulas, N. (2023). Machine learning-based crop stress detection in greenhouses. *Plants*, 12(1), 52.  
<https://doi.org/10.3390/plants12010052>
- Evsahiboğlu, A. N., & Özdikililer, E. (2015). Integration of remote sensing technology to geographic information system for sustainable planning of water resources. *2015 Fourth International Conference on Agro-Geoinformatics (Agro-Geoinformatics)*, 143–148.
- Farhood, H., Bakhshayeshi, I., Pooshideh, M., Rezvani, N., & Beheshti, A. (2022). Recent advances of image processing techniques in agriculture (pp. 129–153).  
<https://doi.org/10.1016/B978-0-323-90508-4.00007-1>
- Gaikwad, N. B., Maheshmalkar, P. R., & Sayyad, S. B. (2025). Vegetation indices for optimizing crop management. *IJFMR – International Journal for Multidisciplinary Research*, 7(3).  
<https://doi.org/10.36948/ijfmr.2025.v07i03.44081>
- Getahun, S., Kefale, H., & Gelaye, Y. (2024). Application of precision agriculture technologies for sustainable crop production and environmental sustainability: A systematic review. *The Scientific World Journal*, 2024(1), 2126734.  
<https://doi.org/10.1155/2024/2126734>
- Ghosh, A., Rai, S., Ashoka, P., Kotyal, K., B, S., Saran, S., Anjali K. P. Sivakumar, Panotra, N., & Pandey, S. K. (2025). Data-driven decision making in agriculture with sensors, satellite imagery and AI analytics by digital farming. *Archives of Current Research International*, 25(5), 37–52.  
<https://doi.org/10.9734/acri/2025/v25i51186>
- Guzel, M., Turan, B., Kadioglu, I., Basturk, A., Sin, B., & Sadeghpour, A. (2024). Deep learning for image-based detection of weeds from emergence to maturity in wheat fields. *Smart Agricultural Technology*, 9, 100552.  
<https://doi.org/10.1016/j.atech.2024.100552>
- Jha, M. K., Chowdary, V. M., & Chowdhury, A. (2010). Groundwater assessment in Salboni Block, West Bengal (India) using remote sensing, geographical information system and multi-criteria decision analysis techniques. *Hydrogeology Journal*, 18(7), 1713–1728.  
<https://doi.org/10.1007/s10040-010-0631-z>
- Jin, X., Li, Z., & Atzberger, C. (2020). Editorial for the special issue “Estimation of crop phenotyping traits using unmanned ground vehicle and unmanned aerial vehicle imagery.” *Remote Sensing*, 12(6), 940.  
<https://doi.org/10.3390/rs12060940>

- Karada, M. S., Bajpai, R., Singh, M., Singh, A. K., Agnihotri, D., & Singh, B. K. (2023). A review on advances in agriculture and agroforestry with GPS and GIS. *International Journal of Plant and Soil Science*, 35(6), 150–160. DOI: 10.9734/IJPSS/2023/v35i62849.
- Khanna, M., Atallah, S. S., Kar, S., Sharma, B., Wu, L., Yu, C., Chowdhary, G., Soman, C., & Guan, K. (2022). Digital transformation for a sustainable agriculture in the United States: Opportunities and challenges. *Agricultural Economics*, 53(6), 924–937. <https://doi.org/10.1111/agec.12733>
- Kumari, K., Mirzakhani Nafchi, A., Mirzaee, S., & Abdalla, A. (2025). AI-driven future farming: Achieving climate-smart and sustainable agriculture. *AgriEngineering*, 7(3), 89. <https://doi.org/10.3390/agriengineering7030089>
- Lei, L., Yang, Q., Yang, L., Shen, T., Wang, R., & Fu, C. (2024). Deep learning implementation of image segmentation in agricultural applications: A comprehensive review. *Artificial Intelligence Review*, 57(6), 149. <https://doi.org/10.1007/s10462-024-10775-6>
- Mălinaş, C. D., Matei, F., Pop, I. D., Sălăgean, T., & Mălinaş, A. (2025). Individual and synergistic contributions of GIS, remote sensing, and AI in advancing climate-resilient agriculture. *AgriEngineering*, 7(7), 230. <https://doi.org/10.3390/agriengineering7070230>
- Mary, S. R., Pachar, S., Srivastava, P. K., Malik, M., Sharma, A., G. Almutiri, T., & Atal, Z. (2022). Deep learning model for the image fusion and accurate classification of remote sensing images. *Computational Intelligence and Neuroscience*, 2022(1), 2668567. <https://doi.org/10.1155/2022/2668567>
- Maurya, D. K., Maurya, S. K., Kumar, M., Chaubey, C., Gupta, D., Patel, K. K., Mehta, A. K., & Yadav, R. (2024). A review on precision agriculture: An evolution and prospect for the future. *International Journal of Plant & Soil Science*, 36(5), 363–374. <https://doi.org/10.9734/ijpss/2024/v36i54534>
- Mohamed, M. (2023). Agricultural sustainability in the age of deep learning: Current trends, challenges, and future trajectories. *Sustainable Machine Intelligence Journal*, 4(2), 120. <https://doi.org/10.61185/SMIJ.2023.44102>.
- Nanasaheb, A. A., V, P. H., & Paswan, P. K. (2020). Soil adjusted vegetation index (SAVI), in the Uben river basin of Gujarat. *International Journal of Chemical Studies*, 8(5), 693–698. <https://doi.org/10.22271/chemi.2020.v8.i5j.10381>
- Nethala, P., Um, D., Vemula, N., Montero, O. F., Lee, K., & Bhandari, M. (2024). Techniques for canopy to organ level plant feature extraction via remote and proximal sensing: A survey and experiments. *Remote Sensing*, 16(23), 4370. <https://doi.org/10.3390/rs16234370>
- Omia, E., Bae, H., Park, E., Kim, M. S., Baek, I., Kabenge, I., & Cho, B.-K. (2023). Remote Sensing in Field Crop Monitoring: A Comprehensive Review of Sensor Systems, Data Analyses and Recent Advances. *Remote Sensing*, 15(2), 354. <https://doi.org/10.3390/rs15020354>.
- Peter, B. G., Messina, J. P., Carroll, J. W., & Chikowo, R. (2021). A case for green-based vegetation indices: Plot-scale sUAS imagery related to crop chlorophyll content on smallholder maize farms in Malawi. *Remote Sensing Letters*, 12(8), 778–787. <https://doi.org/10.1080/2150704X.2021.1938733>
- Raihan, A. (2024). A systematic review of geographic information systems (GIS) in agriculture for evidence-based decision making and sustainability. *Global Sustainability Research*, 3(1), 1–24. <https://doi.org/10.56556/gssr.v3i1.636>
- Singh, S., Sethi, S., Sharma, R., Vaibhavi, D., & Tiwari, U. (2024). Precision agriculture using Web-GIS. *2024 IEEE International Conference on Big Data & Machine Learning (ICBDML)*, 221–226. <https://doi.org/10.1109/ICBDML60909.2024.10577371>
- Srivastava, A., Bharadwaj, S., Dubey, R., Sharma, V. B., & Biswas, S. (2022). Mapping vegetation and measuring the performance of machine learning algorithm in LULC classification in the large area using Sentinel-2 and Landsat-8 datasets of Dehradun as a test case. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLIII-B3-2022*, 529–535. <https://doi.org/10.5194/isprs-archives-XLIII-B3-2022-529-2022>

- Toromade, A. S., & Chiekezie, N. R. (2024). GIS-driven agriculture: Pioneering precision farming and promoting sustainable agricultural practices. *World Journal of Advanced Science and Technology*, 6(1), 057–072.  
<https://doi.org/10.53346/wjast.2024.6.1.0047>
- Upadhyay, A., Chandel, N. S., Singh, K. P., Chakraborty, S. K., Nandede, B. M., Kumar, M., ... & Elbeltagi, A. (2025). Deep learning and computer vision in plant disease detection: A comprehensive review of techniques, models, and trends in precision agriculture. *Artificial Intelligence Review*, 58(3), 92.  
 DOI:10.1007/s10462-024-11100-x.
- V, J. R., & T, T. K. (2025). Application of remote sensing and GIS for real-time crop monitoring and extension support services. *International Journal of Environment and Climate Change*, 15(8), 144–150.
- Won, H. S., Lee, E., Lee, S., Nam, J.-H., Jung, J., Cho, Y., Evert, T., Kan, N., Kim, S., & Kim, D. S. (2025). Image analysis using smartphones: Relationship between leaf color and fresh weight of lettuce under different nutritional treatments. *Frontiers in Plant Science*, 16.  
<https://doi.org/10.3389/fpls.2025.1589825>
- Yi, D., Jun, L., Haodic, G., Xing, Z., Lie, Y., Maidin, S. S., Ishak, W. H. W., & Wider, W. (2024). Transforming agriculture: An insight into decision support systems in precision farming. *Journal of Applied Data Sciences*, 5(4), 1614–1624.  
<https://doi.org/10.47738/jads.v5i4.274>
- Zhang, C., & Kovacs, J. M. (2012). The application of small unmanned aerial systems for precision agriculture: A review. *Precision Agriculture*, 13(6), 693–712.  
<https://doi.org/10.1007/s11119-012-9274-5>

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